Bin Duan, PhD

AI · Computer Vision · Multimodal Learning · Biomedical Image Processing

Pathology Faculty Candidate in **Comparative Disease Computation & AI** · College of Veterinary Medicine

PhD, Computer Science, Illinois Tech, Chicago · **Postdoc**, Cell & Developmental Biology, University of Michigan, Ann Arbor

Research program: Comparative pathology foundation models across species and modalities

Companion animals develop the same spontaneous diseases as people, yet veterinary pathology has almost none of the AI built for human medicine. I build AI that learns one shared representation of disease across species, delivering **predictive**, **multimodal** **foundation-model** methods to veterinary patients first, in tools pathologists use.

THRUST 1
Comparative foundation models
Learning a shared representation, as the predictive engine for different diagnostic tasks.

THRUST 2
Multimodal data integration
Fuse different imaging, molecular, spatial, and clinical modalities into one unified, multimodal patient profile.

THRUST 3
Translation
Deployable, expert-in-the-loop tools for routine clinical or research use in the diagnostic sign-out workflow.

YEARS 1–2
- Canine–human urothelial foundation model
- OnePath pilot in sign-out
- First PhD students recruited
- NSF AI Core / CISE submitted

YEARS 3–4
- Open-weight cross-species foundation model
- Audited multimodal benchmark
- Multiple modalities data fusion
- NIH/NCI and USDA/NIFA proposals

YEAR 5
- Expert-in-the-loop tools validated in clinical sign-out / research routine
- First PhD students graduating
- Offer shared digital pathology services to partner clinics & labs

Collaboration plan

Data curation >> Model development >> Deployment

I lead methods, data, and platform; **partners bring** cases, data, and validation. Expert verification closes the loop.

- **Disease-model partners**: organoids, canine genetics & cancer genomics, human comparator cohorts.
- **Digital-pathology partners**: spatial transcriptomics, AI in the sign-out workflow, cross-species & wildlife breadth.
- **Computational partners**: foundation-model pretraining, computer vision, calibration & statistical rigor.

Fit + funding

Pathology + Institute of AI + Precision One Health

Bridges the imaging-AI gap in comparative disease computation, from foundation model to diagnostic service.

PI-LED **NSF** AI Core · CISE · CAREER · **NIH R21**

COLLABORATIVE **NIH/NCI** (ICDC) · **USDA NIFA** (One Health)

Selected work

- *Towards Saner Deep Image Registration*, **ICCV 2023** Reliable cross-species, cross-modality alignment.
- *Artifact-Minimized High-Ratio Lossy Image Compression with Preserved Analysis Fidelity*, **Nature Communications · under revision** Petascale images accessible without losing analysis fidelity.
- *OnePath*, **research prototype** Open-weight, multimodal, cross-species copilot; the working core of all three thrusts and an early prototype of the diagnostic and research service.

Teaching

Literacy, not just use: students learn **when NOT to trust a model**, running and validating methods against domain shift and noisy labels. Proposed AI-Pathology course **Computational Comparative Pathology**, designed for different levels, contributing to UGA's strategic AI degree programs growth.

Scan and explore

RESEARCH VISION: ▶ OnePath demo · Collaboration plan

PRIOR WORK: Image registration · Image compression · Collaborative system

TEACHING: Course syllabus · Sample lecture slides

DIGITAL PATHOLOGY · **MULTIMODAL DATA INTEGRATION** · **PREDICTIVE MODELS**